

TEEJAS K

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EDUCATION

B.E., Computer Science and Engineering

Regulation 2022 · 2nd Year

PROFILE

Computer Science and Engineering student and independent AI/ML builder. Designs and trains deep learning systems from scratch — spanning 3D generative models, robotics perception, and embedded control — with a focus on production-grade architecture, GPU-efficient training pipelines, and end-to-end deployment.

SKILLS

Python · PyTorch · OpenCV · Modal.com · FastAPI · React · Tailwind CSS · Three.js · ESP32-S3 · FreeRTOS · Raspberry Pi · NVIDIA Jetson · LaTeX · Git · Reinforcement Learning · Computer Vision · LLM Fine-Tuning

PROJECTS

QUADMESH — Industrial Text-to-3D Generation System

~708M params

- Architected an industrial-grade text-to-3D object generation system targeting a single H100 80GB GPU, building on a prior in-house architecture (GLOOMKINS)
- Designed SIVM v2, a four-head structural verification module covering configurable load types, mechanical engineering margins, mesh validity gating, and visual quality scoring
- Authored a full technical paper (LaTeX) documenting architecture, verification pipeline, and explicit surrogate-model caveats
- Building a production frontend (React + Tailwind, dark theme) served via Modal.com, with auto-split for 3D print setup to reduce wasted filament/support material

QUADMESH-CAD — Autonomous CAD Modeling Agent

~2.05B params

- Designed a 2.05B-parameter agentic model dedicated to independently generating, editing, and bug-fixing CAD models
- Split from the core QUADMESH system into a dedicated CAD-focused model, distinct from the general 3D generation model
- Editor operates offline; model runs online only when a generation/edit job is triggered, minimizing GPU cost

ResQ-X6 — Autonomous AI Hexacopter for Disaster Search & Rescue

Pitch Project

- Designed an autonomous hexadrone combining CV-based human detection, GPS victim tagging, emergency payload drop, real-time area mapping, and offline operation
- Trained a multi-hazard detection model from scratch (no pretrained weights) on Modal.com using an H100 80GB GPU, targeting a unified flood/earthquake/wildfire/volcanic dataset
- Hardware: F550 hexacopter frame, DJI Naza-Lite flight controller, Raspberry Pi 4B onboard compute, with inference planned on an NVIDIA Jetson Orin Nano 8GB for full autonomy
- Pitched to investors/incubators with stretch features including gesture detection, victim count estimation, and AI rescue priority scoring